

Data Quality Report (DQR)

Dataset: ppr-group-25208508-train.csv

Module: COMP47350 Group Project – Part (1) DQR

Scope: Data understanding, quality diagnosis, and action recommendations for downstream cleaning and modeling.

1. Executive Summary

This report presents a structured data quality assessment for the training dataset used in the residential property price prediction task. The objective is to identify quality risks that can affect statistical analysis and model robustness, then define practical actions for the next stage (Data Quality Plan, DQP).

The dataset contains **54,000 rows and 9 columns**. At a high level, the data is usable but requires targeted handling in four critical areas:

1. **High missingness** in location/size-related fields (Eircode, Property Size Description).
2. **High-cardinality text features** (Address, Eircode) that are not directly model-friendly.
3. **Price outliers and heavy right skew**, which may destabilize model training.
4. **Small number of exact duplicates** that should be removed for consistency.

Core recommendation: preserve business signal while simplifying fragile raw fields through controlled feature engineering, robust target handling, and transparent step logging in DQP.

2. Dataset Overview

2.1 Shape and Features

- Rows: **54,000**
- Columns: **9**
- Features:
 - Date of Sale (dd/mm/yyyy)
 - Address
 - County
 - Eircode
 - Price (€)
 - Not Full Market Price

- VAT Exclusive
- Description of Property
- Property Size Description

2.2 Time Coverage

Date of Sale is fully populated and valid for all records. Transactions span **2016 to 2024**, with near-even annual counts in this grouped sample.

3. Data Quality Findings

3.1 Missingness

The missingness profile is concentrated in two columns.

Feature	Missing Count	Missing Rate
Property Size Description	51,164	94.75%
Eircode	37,108	68.72%
Other columns	0	0%

Interpretation: - Property Size Description has very limited usable coverage and cannot reliably support size-based modeling decisions in its current form. - Eircode is heavily missing and inconsistent as a direct location key.

3.2 Duplicates and Constants

- Exact duplicate rows: **9** (0.017% of all rows)
- No meaningful fully constant analytical column after preprocessing checks.

Interpretation: - Duplicate count is small but should still be removed to avoid duplicated training signal and metric distortion.

3.3 Cardinality and Feature Usability

High-cardinality fields: - Address: **53,261 unique** values - Eircode: **16,823 unique** values

Moderate/low-cardinality fields: - County: 26 - Description of Property: 3 - Not Full Market Price: 2 - VAT Exclusive: 2

Interpretation: - Raw Address and Eircode should not be directly one-hot encoded. - Structured, low-dimensional derived location features are preferred.

3.4 Price Quality and Distribution

Parsed numeric Price (€) summary: – Mean: **340,639** – Median: **250,000** – Std: **1,662,132** – Min: **5,250** – 5th percentile: **60,000** – 95th percentile: **703,000** – Max: **225,000,000**

Interpretation: – Distribution is strongly right-skewed. – Extreme values (especially very high-end records) can dominate model loss and coefficient estimation. – Robust transformation/clamping should be applied in DQP.

3.5 Category Balance (Key Fields)

County

- Dublin: **16,592** (~30.7%)
- Cork: 5,975
- Kildare: 2,926
- Others with smaller shares

Description of Property

- Second-Hand Dwelling house /Apartment: **44,332**
- New Dwelling house /Apartment: **9,665**
- Minor residual category: 3

VAT and Full-Market Indicators

- VAT Exclusive: Yes = **9,514**, No = 44,486
- Not Full Market Price: Yes = **2,850**, No = 51,150

Interpretation: – Category imbalance is manageable but must be acknowledged in interpretation and model evaluation. – VAT Exclusive carries business-rule implications for target standardization.

4. Visual-Analytical Interpretation (from notebook figures)

The notebook visuals support four consistent conclusions:

1. **Price distribution is heavy-tailed**, and raw boxplots show long upper whiskers and extreme outliers.
2. **County-level heterogeneity is significant**, especially for Dublin/Wicklow compared with many other counties.
3. **Time signal exists** (year-level trend stronger than month-level variation).
4. **High-cardinality location raw fields** are not directly suitable for robust modeling and should be transformed.

These observations justify moving from raw text location fields to engineered location features and from raw price to standardized robust target variants in DQP.

5. Recommended Actions for DQP

5.1 Feature-Level Plan

Feature	DQR Issue	Recommended DQP Action
Price (€)	Heavy skew + extreme outliers	Parse to numeric, apply business-rule standardization and robust clamping
VAT Exclusive + Description of Property	Price definition inconsistency risk	Rule-based VAT adjustment for qualifying new dwellings
Address	Very high cardinality, noisy free text	Extract compact interpretable features, then drop raw address
Eircode	High missingness + high cardinality	Drop or convert only if stable transformed signal exists
Property Size Description	94.75% missing	Drop in final cleaned training set
Date of Sale	Raw date not model-ready	Derive Sale Year and Sale Month
Duplicates	Exact duplicates present	Remove duplicate rows

5.2 Modeling-Safety Principles

- Keep preprocessing deterministic and auditable.
- Avoid leakage from target-derived transformations in predictors.
- Prefer robust target formulation for stable evaluation.
- Ensure no missing values remain in the final DQP output dataset.

6. Risk Assessment

If DQP actions are not applied, expected risks include: 1. Unstable model coefficients due to target outliers. 2. Overfitting and sparse matrices from raw high-cardinality text encoding. 3. Reduced reproducibility from ad-hoc preprocessing. 4. Weaker generalization due to unresolved price-definition inconsistency.

7. Conclusion

The training dataset is adequate for predictive modeling after targeted cleaning and transformation. The DQR confirms that the highest-impact interventions should focus on: –

robust price handling, - location feature engineering, - explicit missingness strategy, - duplicate removal, - and feature-type finalization.

These decisions form the technical bridge from descriptive quality assessment (DQR) to executable cleaning policy (DQP), and they are expected to materially improve model stability and interpretability in later project stages.

Appendix

[Click link to see attached file](#)